

Front End Application for Generating Invoices

The project demonstrates an interface application for creating invoices and generating the created invoice in a PDF format. It is intuitive and interactive. Here is a demonstration of it, accompanied with screenshots.

Start Screen

The screenshot displays the 'Start Screen' of the application, which is used to create and generate invoices. The interface is organized into several sections:

- Date Issued:** A date input field with a calendar icon, showing 'mm/dd/yyyy'.
- Invoice ID:** A text field displaying 'INV-1000'.
- Due Date:** A date input field with a calendar icon, showing 'mm/dd/yyyy'.
- Amount Due:** A text field.
- Billed To:** A section with seven input fields for company details, each with a 'Required' label and a corresponding green button:
 - Company Name
 - Street Number
 - Street Name
 - Postal Code
 - City
 - State (with a note: 'Required, If not applicable, enter N/A')
 - Country
- Reset Billed To Entry:** A button located below the 'Billed To' section.
- Generate Invoice:** A large blue button.
- Purchase Details:** A section with five input fields and corresponding buttons:
 - Product Name
 - Unit Of Measure (with a dropdown arrow)
 - Quantity (with a note: 'Required, for piece or services, only integers')
 - Currency (with a dropdown arrow)
 - Price (with a note: 'Required, max two decimal places')
- Action Buttons:** A row of four buttons at the bottom: 'Add Product', 'Reset Product Entry', 'Reset Product List', and 'Generate PDF'.

Selecting Dates

There is a check so that the date of issue can not be after the due date. In certain cases, it is legal to issue an invoice with a date in the past, therefore, there is no check for whether the date of issue is in the past.

Date of issue and due date can be selected in any order.

If the first selected date is the date of issue and then there is an attempt to select a due date prior to the date of issue, there is a popup notification that this is not allowed and the invalid selection is prevented.

Date Issued
May 21, 2026

Invoice ID
INV-1000

Due Date
mm / dd / yyyy

Amount Due

Billed To

Required **Company Name**

Required **Street Number**

Required **Street Name**

Required **Postal Code**

Required **City**

Required, If not applicable, enter N/A **State**

Required **Country**

Generate Invoice

Purchase Details

Required **Product Name**

Required, Select Unit **Unit Of Measure**

Required, for piece or services, only integers **Quantity**

Required, Select Currency **Currency**

Required, max two decimal places **Price**

Reset Billed To Entry

Add Product **Reset Product Entry** **Reset Product List** **Generate PDF**

And vice versa. If the first selected date is the due date and then there is an attempt to select the date of issue after the due date, again there is a similar popup notification and the invalid selection is prevented.

Date Issued
mm / dd / yyyy

Invoice ID
INV-1000

Due Date
May 21, 2026
mm / dd / yyyy

Amount Due

Billed To

Required **Company Name**

Required **Street Number**

Required **Street Name**

Required **Postal Code**

Required **City**

Required, If not applicable, enter N/A **State**

Required **Country**

Generate Invoice

Purchase Details

Required **Product Name**

Required, Select Unit **Unit Of Measure**

Required, for piece or services, only integers **Quantity**

Required, Select Currency **Currency**

Required, max two decimal places **Price**

Reset Billed To Entry

Add Product **Reset Product Entry** **Reset Product List** **Generate PDF**

Invoice ID

It is generated automatically and with each issued invoice the number is increased with one, for instance: INV-1000 becomes INV-1001 and so on.

Amount Due

It is generated automatically on the basis of the calculation of the total price for all products, including tax. It is the same amount that is on the footer of the list of products in the invoice.

This information is lifted to the top, near the date of issue, the due date, and the invoice number, so that a summary of the most important information can be seen immediately, at a glance. It is especially convenient in cases of invoices with several pages and a long list of products.

Date Issued
May 21, 2026
mm / dd / yyyy

Invoice ID
INV-1000

Due Date
May 28, 2026
mm / dd / yyyy

Amount Due
GBP 779.33

Billed To

Sample Company Name
Company Name

12
Street Number

Sample Street
Street Name

12345
Postal Code

Sample City
City

Sample State
State

Sample Country
Country

Generate Invoice

Purchase Details

second product
Product Name

piece
Unit Of Measure

100
Quantity

GBP
Currency

4.89
Price

Reset Billed To Entry

Add Product

Reset Product Entry

Reset Product List

Generate PDF

No	Product	Unit of Measure	Quantity	Currency	Price	Amount	
1	first product	kilogram	20.21	GBP	10.86	219.48	Delete
2	second product	piece	100.00	GBP	4.89	489.00	Delete

Subtotal

708.48

Tax 10%

70.85

Total

779.33

Billed To

Here, the details of the purchasing company have to be entered. All entry fields must have values, otherwise an invoice can not be generated, respectively, a PDF file can not be generated.

All entry fields in Billed To are marked on the screen as required, and where necessary, there is a short description of the valid entry values.

Button "Reset Billed To Entry"

Clears all the entered information of the purchasing company.
Here is a screen after clicking on the button "Reset Billed To Entry".

Date Issued
May 21, 2026
mm / dd / yyyy

Due Date
May 28, 2026
mm / dd / yyyy

Invoice ID
INV-1000

Amount Due
GBP 779.33

Billed To

Required

Required

Required

Required

Required

Required, If not applicable, enter N/A

Required

Company Name

Street Number

Street Name

Postal Code

City

State

Country

[Reset Billed To Entry](#)

[Generate Invoice](#)

Purchase Details

second product

piece

100

GBP

4.89

Product Name

Unit Of Measure

Quantity

Currency

Price

[Add Product](#)

[Reset Product Entry](#)

[Reset Product List](#)

[Generate PDF](#)

No	Product	Unit of Measure	Quantity	Currency	Price	Amount
1	first product	kilogram	20.21	GBP	10.86	219.48
2	second product	piece	100.00	GBP	4.89	489.00

[Delete](#)

[Delete](#)

Subtotal 708.48

Tax 10% 70.85

Total 779.33

Purchase Details

Here, the details of the products/services being purchased have to be entered. All entry fields must have valid values, otherwise an invoice, or a PDF file, can not be generated.

All entry fields in Purchase Details are marked on the screen as required, and where necessary, there is a short description of the valid entry values.

There are special considerations for the entry fields of Quantity and Price. These are as follows.

Considerations for Quantity in Purchase Details

If the selected unit of measure is piece or services, only integers are allowed. Otherwise, the product can not be added to the invoice list of products.

If the unit of measure is other than the two units mentioned above, the entry can also be a decimal. In this case, there is no restriction for the number of entered decimal places for quantity, though when the product is added to the invoice list the values are rounded to two decimal places.

Depending on the type of business, for instance jewelry, the formula can be reworked to extend the rounding to three, four, or more decimal places, or to drop the rounding altogether.

Considerations for Price in Purchase Details

The entered price must have a maximum of two decimal places. Otherwise, the product can not be added to the invoice list of products.

Invoice List of Products

Each product of the product list can easily be deleted (for instance, in case of a wrong entry) without affecting the other products in the invoice list. Here is the screen, when initially there are two products in the list and the first product is deleted.

The interface displays an invoice form with the following sections:

- Date Issued:** May 21, 2026 (mm / dd / yyyy)
- Invoice ID:** INV-1000
- Due Date:** May 28, 2026 (mm / dd / yyyy)
- Amount Due:** GBP 537.90
- Generate Invoice** button
- Billed To** section with fields for Company Name, Street Number, Street Name, Postal Code, City, State, and Country, each with a corresponding label button.
- Purchase Details** section with fields for Product Name, Unit Of Measure, Quantity, Currency, and Price, each with a corresponding label button.
- Buttons:** Reset Billed To Entry, Add Product, Reset Product Entry, Reset Product List, and Generate PDF.
- Product List Table:**

No	Product	Unit of Measure	Quantity	Currency	Price	Amount	
1	second product	piece	100.00	GBP	4.89	489.00	Delete

- Subtotal:** 489.00
- Tax 10%:** 48.90
- Total:** 537.90

Button "Add Product"

If all entry fields, in Purchase Details, are valid, then upon click it adds product to the list of products, accompanied with a popup notification that the product is added to the list. Here is the screen upon successfully adding a product.

Date Issued

mm / dd / yyyy

Invoice ID

INV-1000

Due Date

May 21, 2026

mm / dd / yyyy

Amount Due

GBP 656.69

Billed To

Sample Company Name

Company Name

12

Street Number

Sample Street

Street Name

12345

Postal Code

Sample City

City

Sample State

State

Sample Country

Country

Reset Billed To Entry

Generate Invoice

Purchase Details

second product

Product Name

piece

Unit Of Measure

100

Quantity

GBP

Currency

Product No 2 added!

Price

Add Product

Reset Product Entry

Reset Product List

Generate PDF

No	Product	Unit of Measure	Quantity	Currency	Price	Amount
1	first product	kilogram	1.56	GBP	10.89	16.99
2	second product	piece	100.00	GBP	5.80	580.00

Subtotal

596.99

Tax 10%

59.70

Total

656.69

Otherwise, there is a popup notification that the product can not be added to the list, due to invalid entries. Here is the screen when the product can not be added, in this case because the unit of measure is piece but the quantity is 1.68.

Date Issued

mm / dd / yyyy

Invoice ID

INV-1000

Due Date

May 21, 2026

mm / dd / yyyy

Amount Due

GBP 18.69

Billed To

Sample Company Name

Company Name

Required

Street Number

Sample Street

Street Name

12345

Postal Code

Sample City

City

Sample State

State

Sample Country

Country

Generate Invoice

Purchase Details

second product

Product Name

piece

Unit Of Measure

1.68

Quantity

GBP

Currency

Price

Can not add product to invoice!

All product fields must have valid values!

Reset Billed To Entry

Add Product

Reset Product Entry

Reset Product List

Generate PDF

No	Product	Unit of Measure	Quantity	Currency	Price	Amount
1	first product	kilogram	1.56	GBP	10.89	16.99

Delete

Subtotal

16.99

Tax 10%

1.70

Total

18.69

Button "Reset Product Entry"

Clears all the entered information of the Purchase Details. It does not affect the list of products. Here is the screen upon clicking on button "Reset Product Entry".

Date Issued
May 21, 2026
mm / dd / yyyy

Invoice ID
INV-1000

Due Date
May 28, 2026
mm / dd / yyyy

Amount Due
2,369.65

Generate Invoice

Purchase Details

Required

Product Name

Required, Select Unit

Unit Of Measure

Required, for piece or services, only integers

Quantity

Required, Select Currency

Currency

Required, max two decimal places

Price

Reset Billed To Entry

Add Product

Reset Product Entry

Reset Product List

Generate PDF

No	Product	Unit of Measure	Quantity	Currency	Price	Amount
1	first product	liter	4.60	GBP	14.18	65.23
2	second product	piece	100.00	GBP	20.89	2,089.00

Subtotal

2,154.23

Tax 10%

215.42

Total

2,369.65

Button "Reset Product List"

Clears all the products in the products list. It does not affect the information in the entry fields of Purchase Details. Here is the screen upon clicking on button "Reset Product List".

Date Issued
May 21, 2026
mm / dd / yyyy

Invoice ID
INV-1000

Due Date
May 28, 2026
mm / dd / yyyy

Amount Due

Generate Invoice

Purchase Details

first product

Product Name

kilogram

Unit Of Measure

10.48

Quantity

GBP

Currency

20.16

Price

Reset Billed To Entry

Add Product

Reset Product Entry

Reset Product List

Generate PDF

No	Product	Unit of Measure	Quantity	Currency	Price	Amount
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Button "Generate PDF"

If all invoice fields are valid, namely, Purchase Details (there is at least one product/service in the list) Billed To, Date Issued, Due Date, Invoice Number, then a PDF is generated, accompanied with a popup notification for this. Here is the screen upon successfully generating PDF.

The screenshot displays a web application for generating invoices. It features several input sections: 'Billed To' with fields for company name, street, postal code, city, state, and country; 'Purchase Details' with fields for product name, unit of measure, quantity, currency, and price. Summary fields at the top right show 'Date Issued' (May 21, 2026), 'Invoice ID' (INV-1000), 'Due Date' (May 28, 2026), and 'Amount Due' (GBP 1,134.24). A 'Generate Invoice' button is present. Below the input fields are buttons for 'Reset Billed To Entry', 'Add Product', 'Reset Product Entry', 'Reset Product List', and 'Generate PDF'. A table lists the products added: 'first product' (1.26 kg, GBP 2.48) and 'second product' (100.00 pieces, GBP 1,028.00). A summary table at the bottom right shows 'Subtotal' (1,031.12), 'Tax 10%' (103.11), and 'Total' (1,134.24). A green notification bubble in the bottom right corner states 'PDF generated!'.

No	Product	Unit of Measure	Quantity	Currency	Price	Amount
1	first product	kilogram	1.26	GBP	2.48	3.12
2	second product	piece	100.00	GBP	10.28	1,028.00

Subtotal	1,031.12
Tax 10%	103.11
Total	1,134.24

Otherwise, there is another popup notification that the PDF can not be generated due to invalid invoice entries. Here is the screen when a PDF file can not be generated, in this case because date of issue is not entered.

Date Issued

mm / dd / yyyy

Invoice ID

INV-1000

Due Date

May 28, 2026

mm / dd / yyyy

Amount Due

GBP 1,426.11

Billed To

Sample Company Name

Company Name

12

Street Number

Sample Street

Street Name

12345

Postal Code

Sample City

City

Sample State

State

Sample Country

Country

Generate Invoice

Purchase Details

second product

Product Name

piece

Unit Of Measure

100

Quantity

GBP

Currency

10.89

Price

Can not generate PDF!

All invoice fields must have valid values!

Reset Billed To Entry

Add Product

Reset Product Entry

Reset Product List

Generate PDF

No	Product	Unit of Measure	Quantity	Currency	Price	Amount	
1	first product	kilogram	10.24	GBP	20.26	207.46	Delete
2	second product	piece	100.00	GBP	10.89	1,089.00	Delete

Subtotal

1,296.46

Tax 10%

129.65

Total

1,426.11

Button "Generate Invoice"

If all invoice fields are valid, namely, Purchase Details (there is at least one product/service in the list), Billed To, Date Issued, Due Date, Invoice Number, then an invoice is generated, accompanied with a popup notification for this. The invoice number is increased with one, for instance, from INV-1000 to INV-1001. Here is the screen upon successfully generating an invoice.

Date Issued
May 21, 2026
mm / dd / yyyy

Invoice ID
INV-1000

Due Date
May 28, 2026
mm / dd / yyyy

Amount Due
GBP 1,426.11

Invoice INV-1000 generated!

Generate Invoice

Purchase Details

second product

Product Name

piece

Unit Of Measure

100

Quantity

GBP

Currency

10.89

Price

Reset Billed To Entry

Add Product

Reset Product Entry

Reset Product List

Generate PDF

No	Product	Unit of Measure	Quantity	Currency	Price	Amount	
1	first product	kilogram	10.24	GBP	20.26	207.46	Delete
2	second product	piece	100.00	GBP	10.89	1,089.00	Delete

Subtotal

1,296.46

Tax 10%

129.65

Total

1,426.11

Billed To

Sample Company Name

Company Name

12

Street Number

Sample Street

Street Name

12345

Postal Code

Sample City

City

Sample State

State

Sample Country

Country

Otherwise, there is another popup notification that the invoice can not be generated due to invalid invoice entries. Here is the screen when an invoice can not be generated, in this case because there is no entry for street number in Billed To.

Billed To

Sample Company Name

Company Name

Required

Street Number

Sample Street

Street Name

12345

Postal Code

Sample City

City

Sample State

State

Sample Country

Country

Reset Billed To Entry

Date Issued
May 21, 2026
mm / dd / yyyy

Due Date
May 28, 2026
mm / dd / yyyy

Invoice ID
INV-1001

Amount Due
GBP

Can not generate Invoice!
 All invoice fields must have values!

Generate Invoice

Purchase Details

second product

piece

100

GBP

10.89

Product Name

Unit Of Measure

Quantity

Currency

Price

Add Product

Reset Product Entry

Reset Product List

Generate PDF

No	Product	Unit of Measure	Quantity	Currency	Price	Amount
1	first product	kilogram	10.24	GBP	20.26	207.46
2	second product	piece	100.00	GBP	10.89	1,089.00

Subtotal

1,296.46

Tax 10%

129.65

Total

1,426.11

The presented application handles the front end and the invoice generation is simulated. The invoice is not added to a database, only the number of the invoice is increased. Button "Generate Invoice" can serve as the entry point, the trigger, towards extending the application to the back end and adding the invoice to a database.

Overview of the generated PDF file

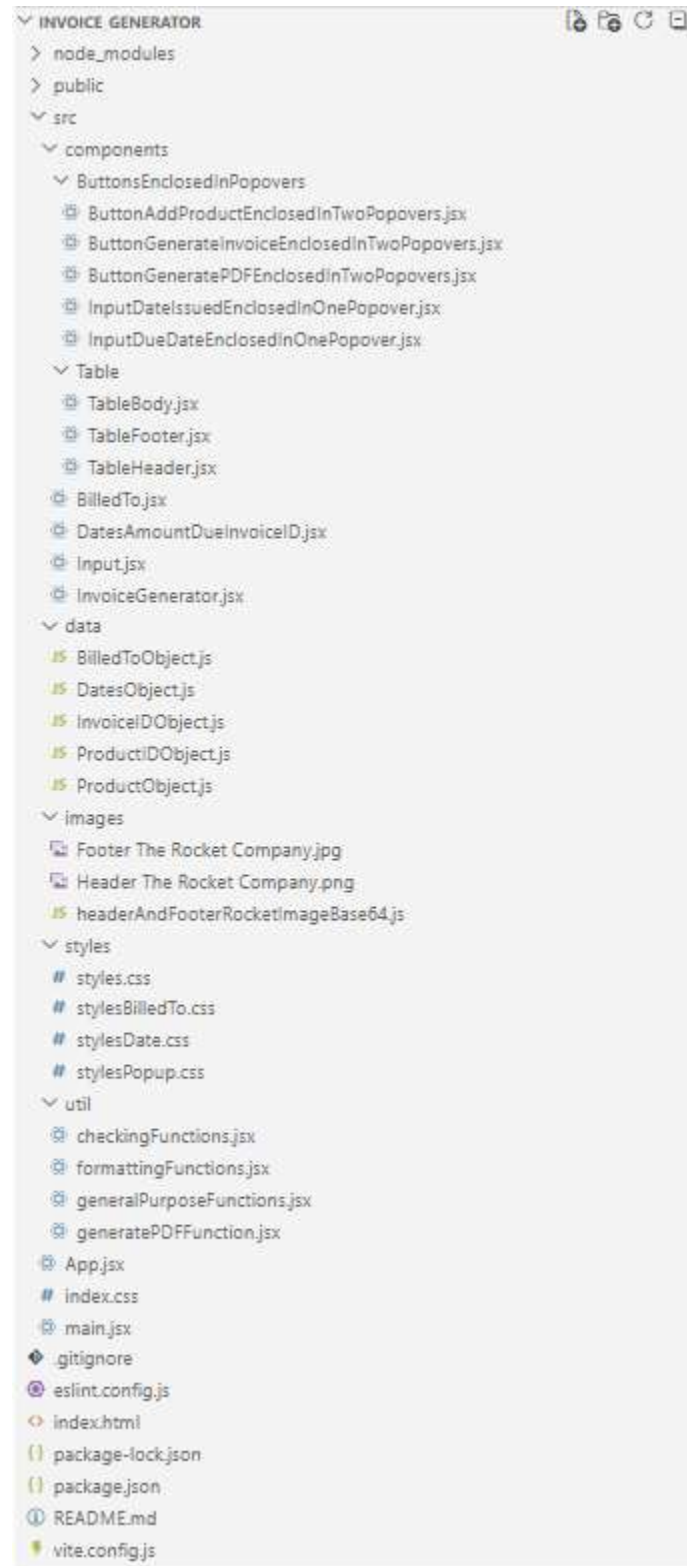
The generated PDF file for the invoice has a fictitious company name, logo, address, bank account, tax number, etc., of the invoice-issuing company, just to present how it would look like in the real world.

Each page of the invoice is numbered and there is a footer on each page, where the logo, the name of the company, its tax number and bank account are displayed in small fonts.

These, of course, can be reworked for whatever is required by the company that uses this invoice.

File "Sample Invoice in PDF format", as the name suggests, contains a sample invoice. It consists of two pages.

Project Structure Overview



The entry React component, one level above App.jsx, is InvoiceGenerator.jsx

The names of the React components, methods, fields, etc, are self-explanatory and, where necessary, accompanied with comments. When it comes to the names of objects, the word "Object" is added to the name as a suffix, in order to differentiate immediately whether it is a React component or not.

Thus, for example, React component BilledTo can be differentiated at first glance, without further inspection, from object BilledTo because the latter has the suffix Object to its name and is specified as BilledToObject.

Folder "components": As the name suggests, it contains React components.
It also has two subfolders.

Subfolder "ButtonsEnclosedInPopovers": contains Popovers, wrapped around Buttons.

Subfolder "Table": contains the table for the list of products.

Folder "data": contains the objects applied in the project.

Folder "images": contains the images that are applied as header and footer.

Folder "styles": contains the files for the CSS formatting.

Folder "util": contains helper functions applied in the project.